

DocuMind Research Paper: Advances in Retrieval-Augmented Generation  
Author: AI Systems Lab (September 2026)

Abstract:

Retrieval-Augmented Generation (RAG) significantly reduces factual hallucinations in large language models by grounding output in domain-specific corpus excerpts. In this work, we demonstrate that chunking with 700 characters and 120 character sliding-window overlap preserves semantic continuity across complex paragraphs. Experiments indicate an accuracy boost from 64.2% to 91.8% on factual benchmarks.

Section 1: Architecture & Page Grounding

Standard LLMs often suffer from catastrophic forgetting and temporal cutoff. By binding page numbers and chunk offsets directly to vector embeddings, the inference engine provides users with verifiable citations for all factual claims.

## Section 2: High-Dimensional Dense Vector Embeddings

We evaluate text-embedding-004 generating 768-dimensional normalized unit vectors.

Cosine similarity measures the orientation between query vectors and document chunk vectors.

When combined with lexical keyword weighting (BM25-lite bonus of 0.15), retrieval precision increases by 14.3% compared to pure dense search alone.

### Key Finding on Page 2:

The hybrid retrieval strategy eliminates false positive matches on technical acronyms.

Total indexing latency per 100-page document remains under 4.2 seconds.

### Section 3: Benchmark Evaluations and Conclusions

We tested DocuMind RAG on a dataset of 5,000 regulatory compliance and medical documents.

#### Key Performance Results:

1. Precision@5 reached 94.7% with sub-second retrieval latency (180ms).
2. Exact page attribution accuracy was verified at 99.1%.
3. Interactive multi-choice quiz generation scored 92% comprehension among test cohorts.

#### Conclusion & Operational Recommendations:

1. Always index documents with page-level metadata rather than raw continuous text streams.
2. Implement citation badges directly in chat UIs to build user trust and transparency.
3. Provide side-by-side comparative analysis when evaluating revisions or multi-source reports.